

# ML Evaluation, Metrics, and Experiments

Evaluation answers a specific question about generalization and value. Accuracy, precision, recall, ranking, calibration, and experiment outcomes measure different things. Choose the metric and data split from the decision, quantify uncertainty, and investigate failure slices rather than chasing a single leaderboard number.

**For:** Candidates who can train a model but need to explain whether it is useful, whether the evaluation is valid, and how offline results relate to online decisions.

Suggested pace: 2-3 weeks of worked metric exercises | Basic supervised learning

## What you will be able to do

- Compute and interpret classification metrics under imbalance.
- Distinguish ranking quality, calibrated probabilities, and threshold decisions.
- Design valid temporal or grouped evaluation and avoid test-set tuning.
- Explain A/B-test assumptions, guardrails, and offline/online disagreements.

## Build the right mental model

### The denominator changes the meaning

Precision asks how many predicted positives are truly positive; recall asks how many actual positives were found. An all-negative classifier can have high accuracy when events are rare. Report the confusion matrix and operating point, not a metric name alone.

### Ranking and calibration are different

A model can order examples well but produce probabilities that are too high or low. Threshold and expected-cost decisions may need calibration and validation. AUC does not directly tell you the operational false-positive volume at a chosen cutoff.

### Experiments need an analysis plan

Specify the unit of randomization, primary metric, guardrails, expected effect, and observation window before launch. Repeated unplanned peeking and selective reporting can produce misleading confidence. Account for delayed outcomes and incomplete cohorts.

# Study with evidence.

Work through small numeric examples by hand, then verify with a library. Keep data split, threshold, and metric choices separate. When comparing models, use the same valid evaluation population and report uncertainty and important error slices.

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## Sessions 1-3: Classification metrics

Build confusion matrices and compute accuracy, precision, recall, and F1. Vary prevalence and threshold. Translate rates into absolute review volume and business cost using an explicitly hypothetical workload.

**Deliverable and check:** A numeric comparison showing why high accuracy can coexist with poor rare-event detection.

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## Sessions 4-6: Ranking and calibration

Study ROC and precision-recall curves, top-k metrics, calibration, and threshold selection. Separate a score used to rank from a probability used for an economic decision.

**Deliverable and check:** A justified operating point and an explanation of what the chosen metric does not prove.

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## Sessions 7-10: Valid splits and slices

Choose temporal, grouped, or random splits according to deployment. Test leakage and feature timing. Compare cohorts and report sample size and uncertainty so a noisy slice is not mistaken for a certain effect.

**Deliverable and check:** An evaluation report with a baseline, mature labels, and a defensible comparison population.

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## Sessions 11-14: Online experimentation

Define an A/B-test plan with assignment, guardrails, duration rationale, and rollback criteria. Investigate a hypothetical offline win that loses online through latency, feature skew, or objective mismatch.

**Deliverable and check:** A decision memo that separates evidence, uncertainty, and proposed follow-up experiments.

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### A repeatable practice session

Spend about 15 minutes recalling the previous topic without notes, 45 minutes on today's problem or lab, and 15 minutes explaining and checking the result. Record the failed assumption, not just the final answer. Revisit weak material after a few days and again the following week.

# Apply it. Explain it.

## Evaluate a rare-event classifier at a real operating point

Among 10,000 examples, 100 are positive. A classifier finds 60 positives, misses 40, and raises 90 false alarms. You must explain quality and the workload created for a human review team.

### A worked approach

1. The confusion matrix is  $TP=60$ ,  $FN=40$ ,  $FP=90$ , and  $TN=9,810$ . Accuracy is  $9,870/10,000 = 98.7\%$ , which is lower than the 99% all-negative baseline even though the classifier finds useful positives.
2. Precision is  $60/150 = 40\%$ ; recall is  $60/100 = 60\%$ ; F1 is 0.48. The review team receives 150 alerts, of which 90 are false alarms. Whether this is useful depends on review capacity and error costs.
3. Choose thresholds using validation data and compare alert volume, precision, recall, and cost. Keep the final test set isolated. If the team can review only 80 alerts, evaluate the appropriate top-80 or threshold-constrained policy rather than reporting unconstrained recall.

#### Expected result

The numbers show why no single metric decides usefulness. A defensible evaluation connects ranking, threshold, operational capacity, and the consequence of missed or false alerts.

### Verify the result

- Do all four confusion-matrix cells sum to 10,000?
- Is the threshold chosen without repeated test-set tuning?
- Are review capacity and label maturity represented?

### Extend the exercise

After deployment, labels arrive weeks later. Define how monitoring distinguishes preliminary counts from mature outcomes and avoids declaring a model better because recent negatives have not yet had time to become positive.

# Make the learning stick.

## Does a confidence interval containing no change prove equivalence?

No. It may mean the experiment lacks evidence for a directional effect at that precision. Equivalence requires its own margin and analysis. Consider sample size, effect size, and the pre-specified decision rule.

## Why might offline lift disappear online?

Possible causes include biased data, different exposure, feature delay, model/version mismatch, latency, and a metric that poorly reflects the product objective. Diagnose the path rather than assuming either evaluation is automatically wrong.

## Should I report every slice as a separate significant finding?

No. Small samples and multiple comparisons can create noisy apparent differences. Use pre-specified important cohorts where possible, report uncertainty, and investigate rather than overclaiming exploratory patterns.

## Avoid these traps

- Selecting thresholds repeatedly on the final test set.
- Reporting accuracy alone for a rare-event task.
- Ignoring delayed labels or changing evaluation populations.

## Ready for the next step?

- Metrics can be derived from the confusion matrix.
- The operating point reflects real decision constraints.
- Splits, labels, and uncertainty are documented.
- Offline and online evidence are interpreted within their scope.

## Reliable next reads

- [scikit-learn model evaluation](#)
- [Google ML classification](#)
- [scikit-learn common pitfalls](#)

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